Distribution (Mods S.O.O. 6004) – General Data

1. General

Refer to Figure 1 for details.

The pneumatic package is located on the left side of the cabin roof at station 177.00 approximately. The pressure supply for the pneumatic package is tapped from the bleed air system line on the left side of the cabin roof. (Refer to 21-00-00, Air Conditioning – General Data).

A. Pneumatic Package

The pneumatic package consists of a heat exchanger, a pressure regulator, a strainer, low pressure switch, and connecting pipe lines.

B. Heat Exchanger

The heat exchanger is a rearward facing airscoop mounted on top of the fuselage at station 176.50. The double walls of the airscoop and the internal baffle system form the heat exchanger. Bleed air enters the heat exchanger through the outboard pipe, and is directed through the baffle system which exposes the air to the entire surface area of the inner and outer walls of the scoop, where cooling air passing over these surfaces dissipates the heat from the bleed air. The cooled air leaves the heat exchanger through the inboard pipe.

C. Pressure Regulator

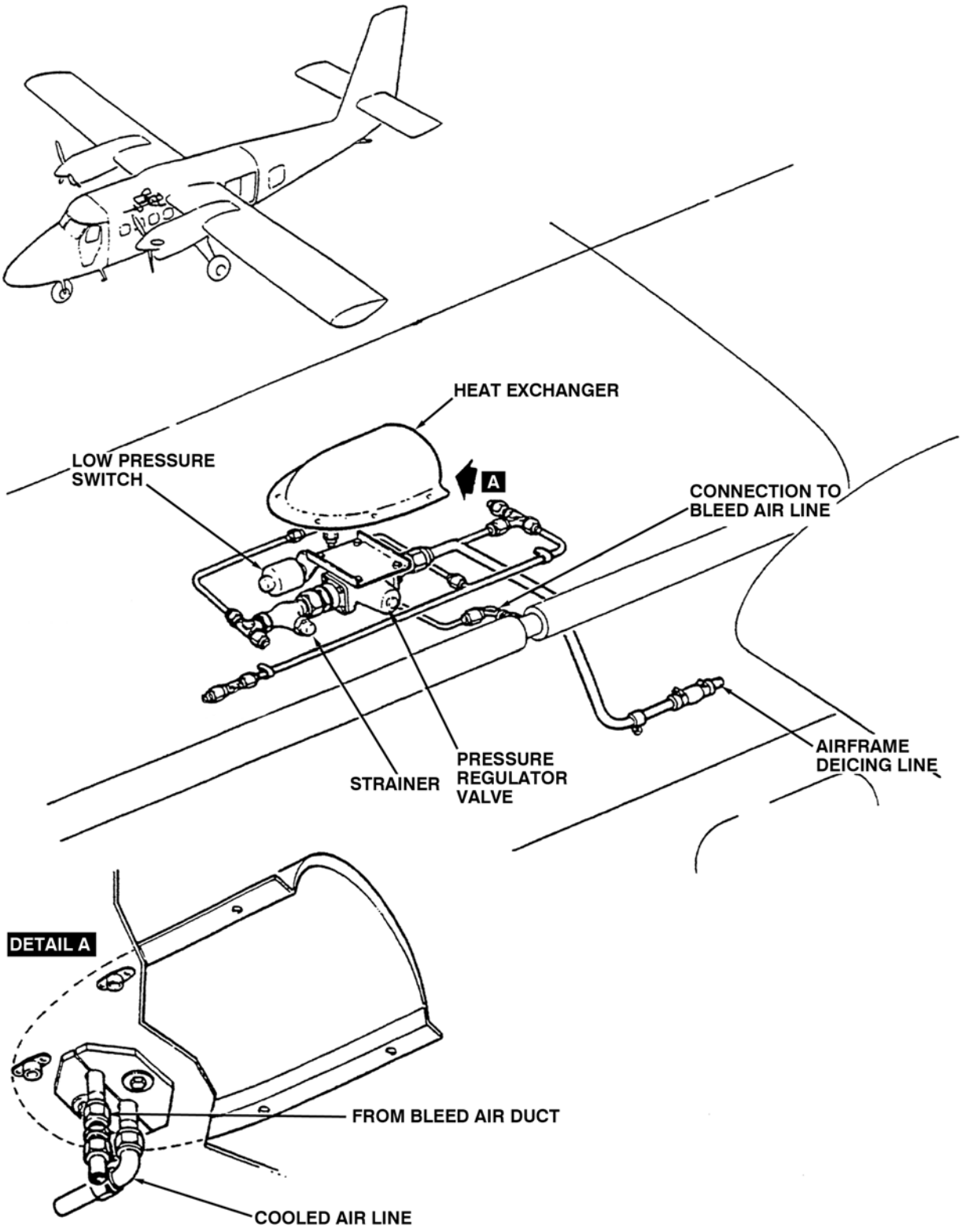
The pressure regulator is mounted on a plate which is secured to the cabin roof. It is set to regulate the pneumatic system at 18 psi. A safety relief valve on the pressure regulator is set to relieve pressure at 25 psi should the regulator section fail.

D. Strainer

The strainer incorporates a 60–mesh wire element and is fitted to the intake port of the pressure regulator in the cooled air line from the heat exchanger to prevent impurities in the bleed air from entering the pneumatic system.

E. Low Pressure Switch

The low pressure switch is set to illuminate a caution CAS message should the system pressure drop to between 13 and 15 psi. When pressure increases to between 16 and 18 psi, the switch contacts open to extinguish caution CAS message. If the aircraft is on the ground, a pneumatic pressure low advisory message will activate.



Pneumatic Package
Figure 1

2. Servicing

A. Strainer

1) Remove cabin ceiling panels. (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data).

2) Detach upholstery in cabin roof.

3) Cut lockwire and unscrew strainer. Remove wire element.

4) Clean wire element and strainer.

5) Position wire element in strainer, install strainer and secure with lockwire.

6) Install cabin ceiling panels. (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data).

3. Adjustment/Test

A. Function Test Pneumatic Package

Refer to Figure 1 for details.

Equipment required:

- Source of compressed air 100 psi
- Pressure gauge 0 to 30 psi
- Pressure test rig consisting of a pressure tank capable of withstanding 160 psi and having a volume of 10 cu ft, a 0 to 100 psi pressure gauge, and two shut-off valves to isolate the rig from the aircraft and the compressed air source

1) Remove cabin ceiling panels (refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data) in location of station 177.00 to provide access to heat exchanger.

2) Disconnect bleed air line at inlet to heat exchanger. Cap end of line, and connect pressure rig to union on heat exchanger.

3) On aircraft with airframe de-icing, disconnect the pressure supply line at the ejector located in the rear fuselage, station 468.00. Cap the fitting on ejector.

4) Carry out pressure test as follows:

a) Remove cap on pneumatic package and install 0 to 30 psi pressure gauge.

b) Set pressure rig to 80 psi. Check that low pressure caution CAS message goes out when pressure is between 16 and 18 psi. Check pressure gauge at forward tee reads 18 ± 2 psi. Adjust regulator valve to obtain correct pressure, if necessary.

c) Lower rig pressure gradually to 25 psi. Check that reading on pressure gauge is maintained at 18 ± 2 psi.

d) Lower rig pressure gradually and check that low pressure caution CAS message illuminates when pressure gauge reads between 13 and 15 psi.

e) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

f) Shut off rig pressure. Remove pressure gauge from forward tee and install cap.

g) Disconnect pressure rig. Connect bleed air line to union on heat exchanger.

h) Install cabin ceiling panels. (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data).

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